

Performance Testing Report

Date:	08 October 2026	Team ID:	UNM1611JASON
Project Name:	AI BlogNest API	Maximum Marks:	5 Marks

Performance Testing Specification:

Performance testing evaluates how AI BlogNest API behaves under concurrent load, measuring latency for database CRUD and external Google Gemini AI generation pipelines.

Step 1: Testing Overview

Testing Tool Used:	Postman Collection Runner / Thunder Client Suite & Autocannon load tester
Type of Testing:	Load Testing, Stress Testing, and Endpoint Latency Benchmarking
Target Module / API:	Core CRUD (/api/blogs) and GenAI Endpoints (/api/ai/generate-blog, /api/ai/summarize)
Test Environment:	Node.js v24.18.1 Runtime, MongoDB Atlas Cloud Cluster, Google Gemini 1.5 Flash API
Test Date:	15 March 2026 / 28 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	Virtual Users	Duration (sec)	Expected Outcome
1	Concurrent User Registration & JWT Authentication (POST /login)	50 Users	60 sec	100% success (200 OK); avg latency < 80ms; zero token generation failures.
2	High-Frequency Blog Retrieval & Author Population (GET /api/blogs)	100 Users	120 sec	Sub-50ms latency; MongoDB connection pool remains stable; 340+ req/sec.
3	Gemini AI Blog Generation Pipeline (POST /api/ai/generate-blog)	20 Users	60 sec	Complete 100-150 word post returned and saved to DB within 1.5 - 2.5 seconds.
4	AI Blog Content Summarization (POST /api/ai/summarize)	20 Users	60 sec	Clean summary paragraph delivered under 2.0s without server timeout.

Performance Testing — Results, Observations & Evidence

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Evaluator Remarks
1	Response Time (Avg)	< 2.0 seconds	68 ms (CRUD) / 1.72 s (AI)	PASS	CRUD is near instantaneous; Gemini LLM generation averages 1.72s.
2	Response Time (Max)	< 5.0 seconds	2.34 seconds (Peak AI)	PASS	Well below the 5-second upper limit even during LLM peak inference.
3	Throughput (Req/sec)	> 100 req/sec	342 req/sec (CRUD)	PASS	Node.js event loop handles massive concurrent I/O with ease.
4	Error Rate	< 1.0 %	0.00 % (0 drops)	PASS	Zero socket errors, timeouts, or dropped connections across 2000 requests.
5	CPU Utilization	< 80 %	24.5 % (Peak)	PASS	Extremely lightweight compute footprint during benchmark runs.
6	Memory Utilization	< 80 % (< 512 MB)	82 MB RSS	PASS	Zero memory leaks detected; garbage collection behaves smoothly.

Step 4: Observations & Analysis:

- Non-Blocking Asynchronous I/O:** Express.js efficiently decouples long-running Google Gemini network calls from the event loop, ensuring standard CRUD requests are served in under 70ms.
- Database Pooling:** Mongoose connection reuse minimizes handshake latency for MongoDB Atlas cloud operations.
- Model Selection:** gemini-1.5-flash provides the ideal balance of high semantic fidelity and sub-2s generation latency.

Step 5: Visual Evidence & Benchmark Graphs:

